

Team Charter

Quantum Routines: Implementation, Verification, and Resource Estimation

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Industry Mentor: Mariia Mykhailova (PsiQuantum)

Expertise and Growth (Needs finishing)

Kush Aggarwal —

Thomas Huang —

Christopher Sharp — My expertise is in optimization e.g. through multi-objective cost functions, which could be applied to estimating and improving resources of quantum routines. I am familiar with Python and have some background in lower level classical computing. I think I could improve on both of those. I also haven't meaningfully contributed to a larger project yet, and while I am familiar with the concepts of CI/CD I haven't implemented them much in practice before. Finally, by understanding and implementing at least one quantum algorithm into Workbench, I hope to build a deeper understanding of reversible computing and quantum programming.

Support Needs (Needs improvement)

We could use help further understanding Workbench and how to implement algorithms in a continuous integration process.

Team Rules and Expectations (Needs improvement)

- How will we make sure everyone is heard?
 - A. Everyone should feel free to voice their opinion on a matter and we will try to solicit the viewpoints of team members who have been silent in case they hadn't felt comfortable speaking up.
- How will we resolve team conflicts?
 - B. We will attempt to resolve conflicts through communication. If there remains an issue, we will discuss with our industry mentor and/or our instructor for mediation.

Communication Guidelines (Improvement?)

- We will meet twice a week with our industry mentor and with the instructor. The former meeting will be remote. The latter meeting will be mixed remote and in-person as needed.
- We will use email and (as needed) GitHub Issues to communicate directly in between meetings.